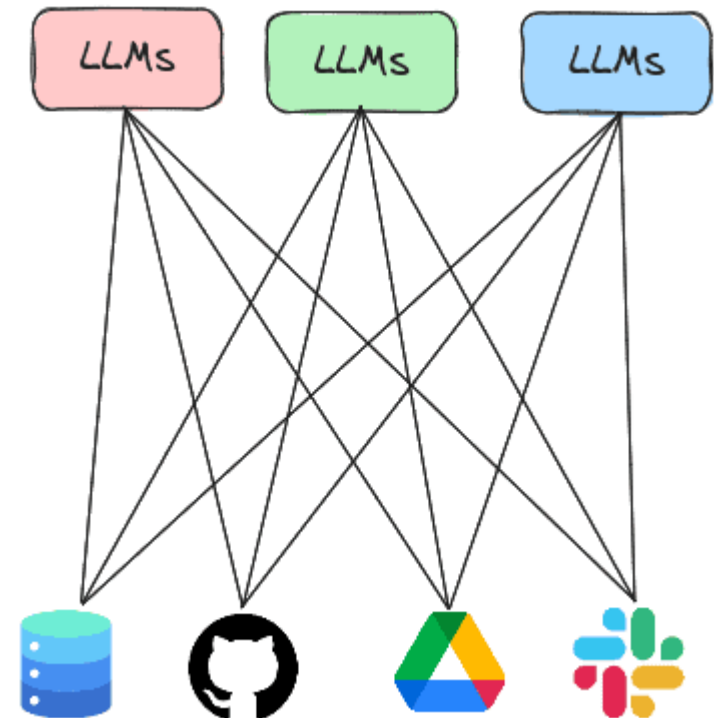


Model Context Protocol (MCP)

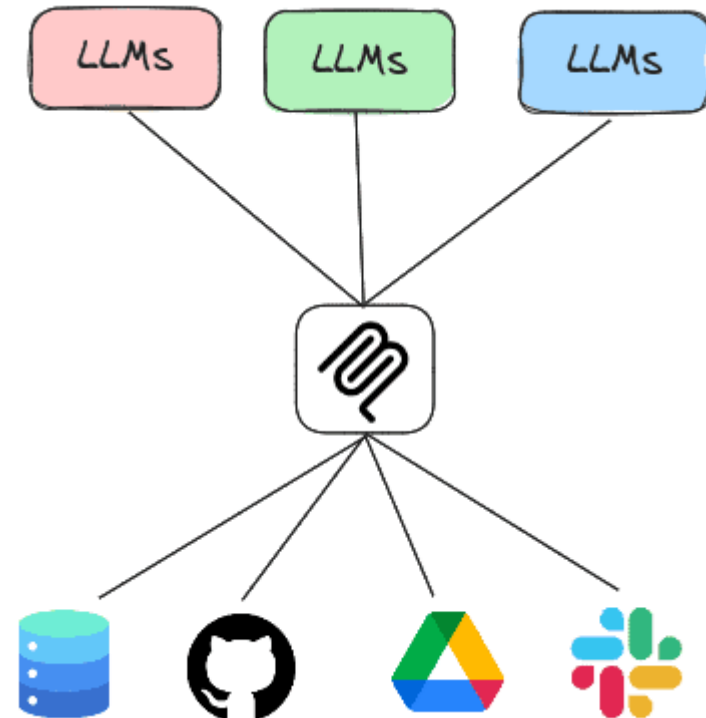
Motivation

- Recall prior agents
 - Allow LLMs to execute built-in tools or custom tools implemented by the application developer to handle user query
- Implemented within application
 - But, many different implementations
 - Fragmented tool development
 - Individual applications must change as services change



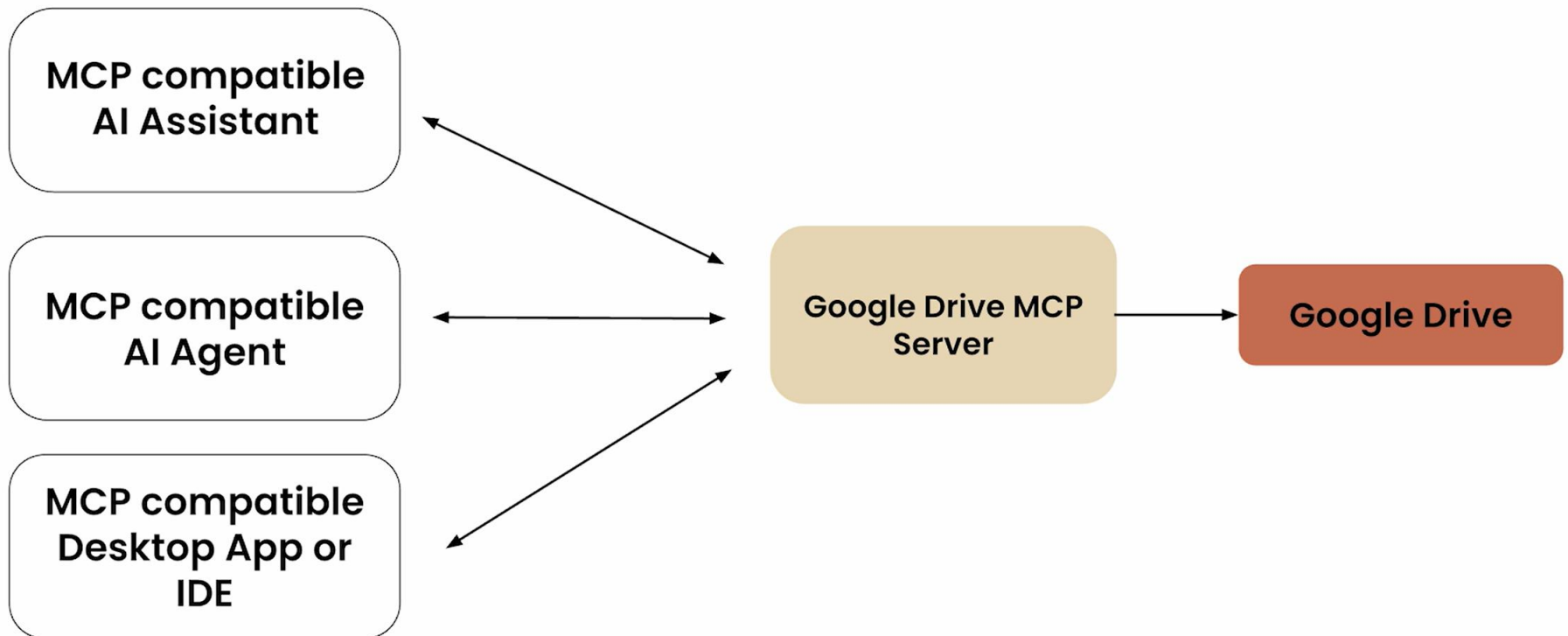
Model Context Protocol (MCP)

- Anthropic (11/2024)
- Standard open-source approach for connecting models to tools
 - Done via JSON-RPC messaging
- Enables LLM application to invoke tools provided remotely
- Analogy
 - Browser or app discovering and accessing web services via HTTP
 - LLM agent discovering and invoking tools via MCP
- Effectively an interoperability layer for connecting LLM agents to tool functionality (both local and remote)



Example

- Google Drive MCP server re-usable by a variety of AI applications



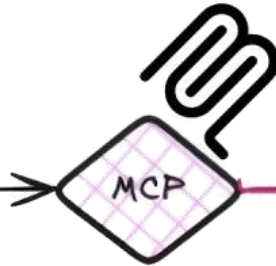
Many MCP reference servers

- <https://github.com/modelcontextprotocol/servers>

AI models and agents



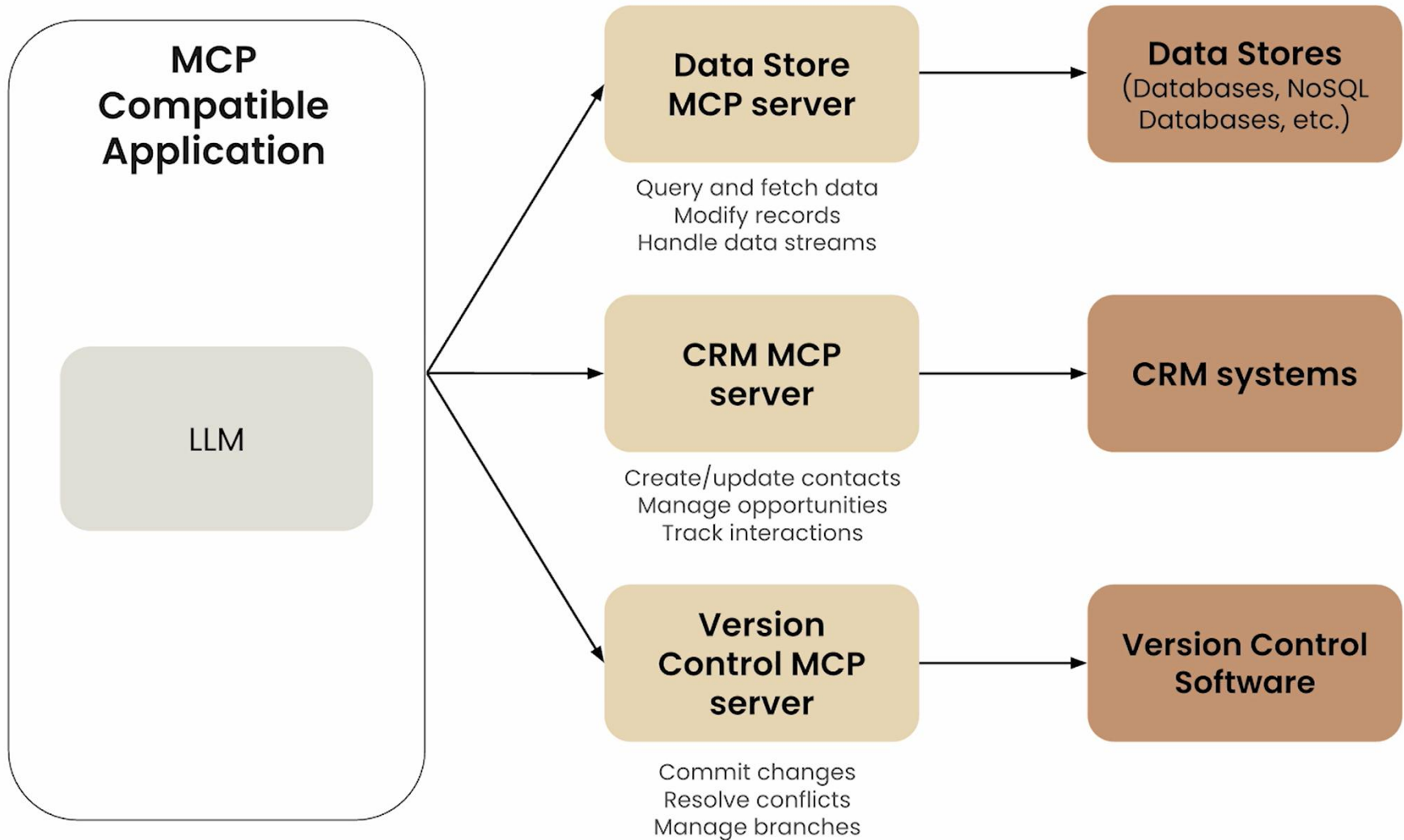
Model Context Protocol

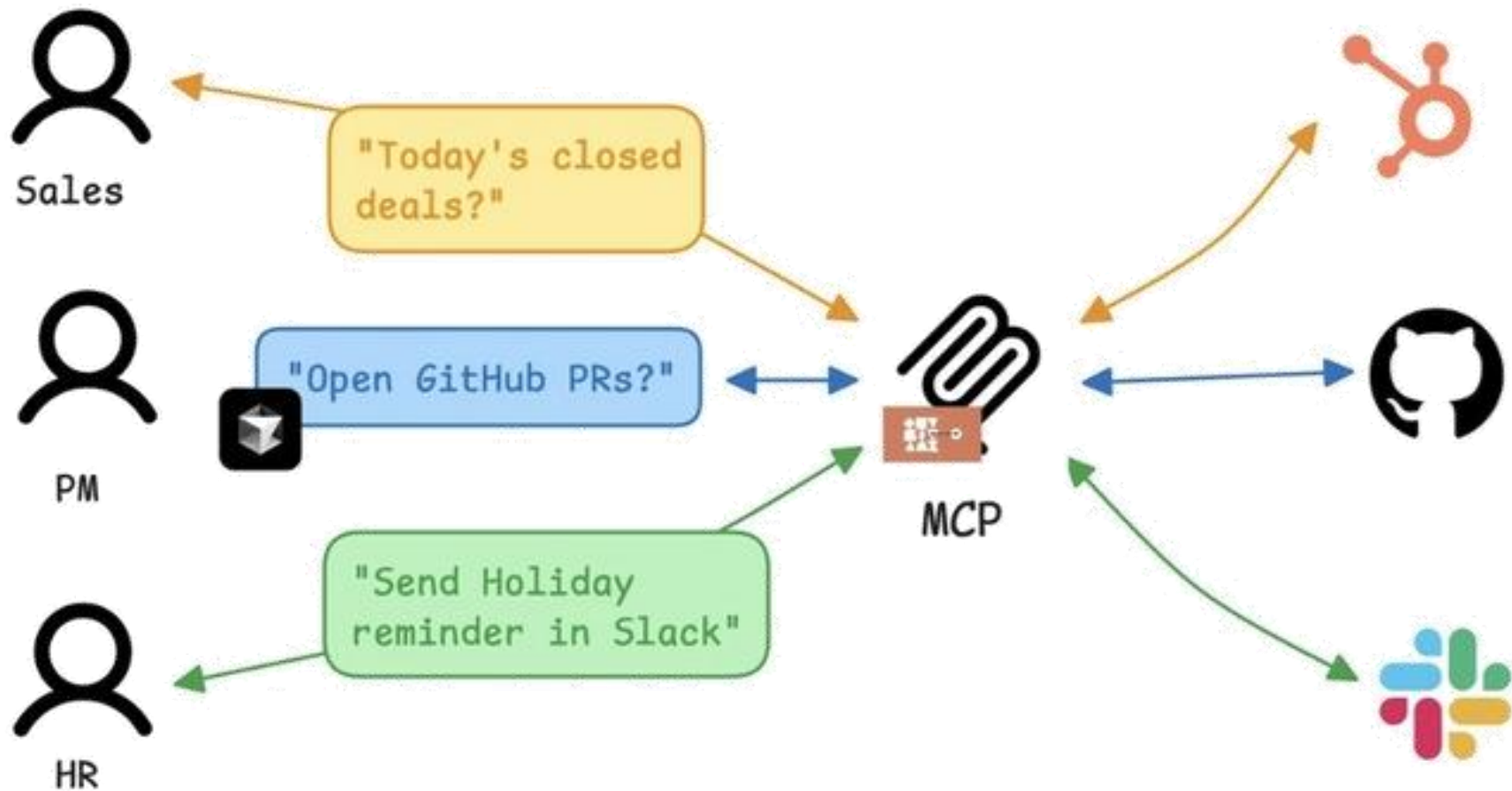


Tools and data



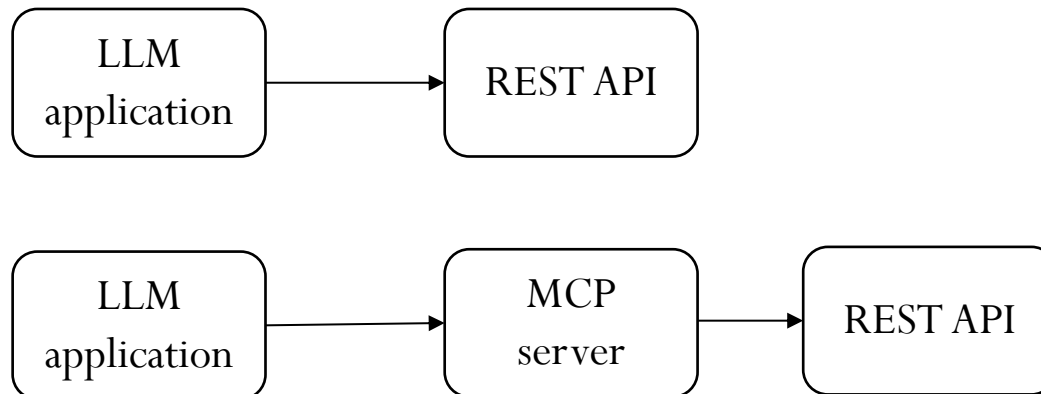
With MCP: Standardized AI Development





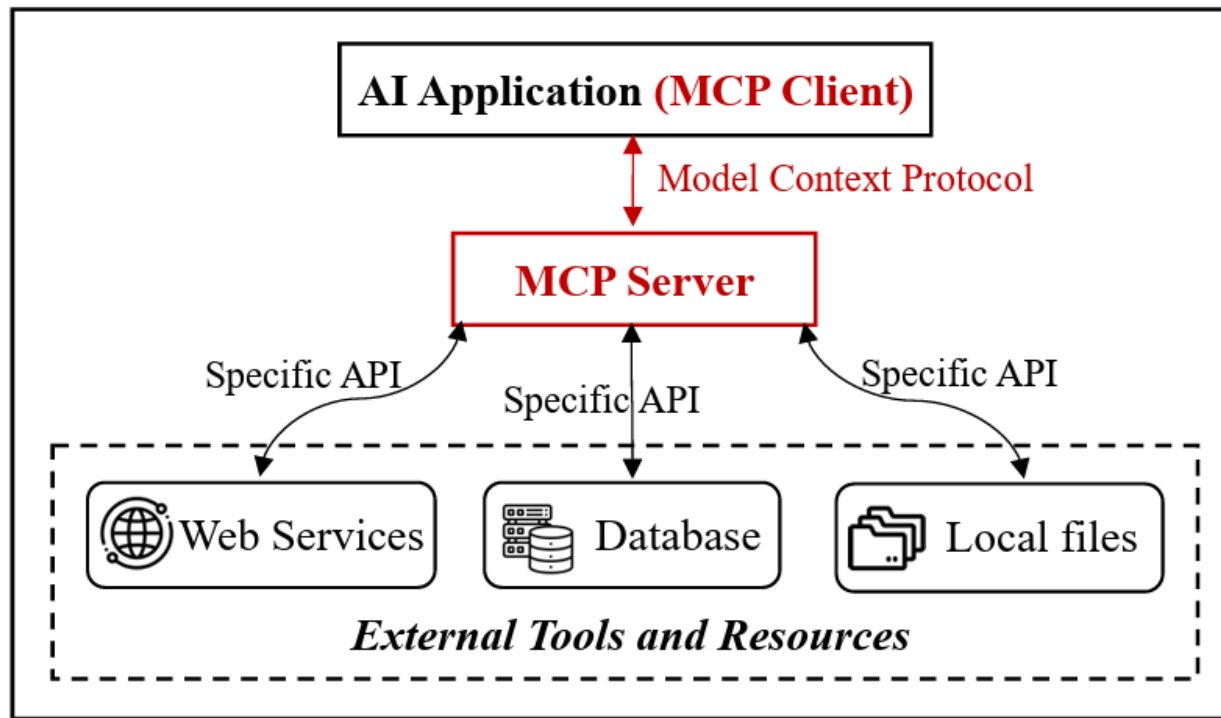
MCP vs REST APIs

- Analogous to REST APIs
 - Standardized way for web clients to interact with backend web services
- MCP
 - Standardized way for AI applications to interact with external data and services
- Can go from...

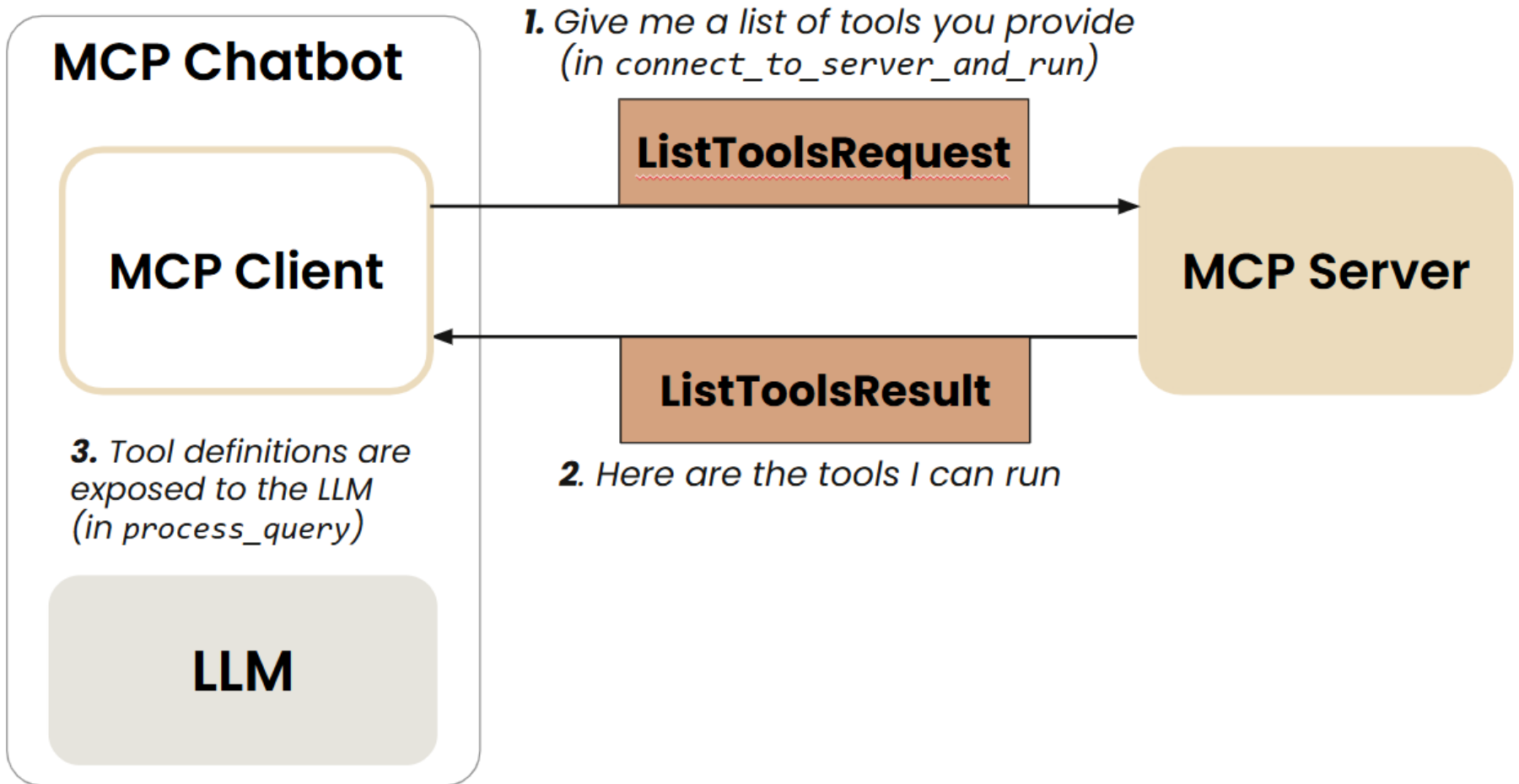


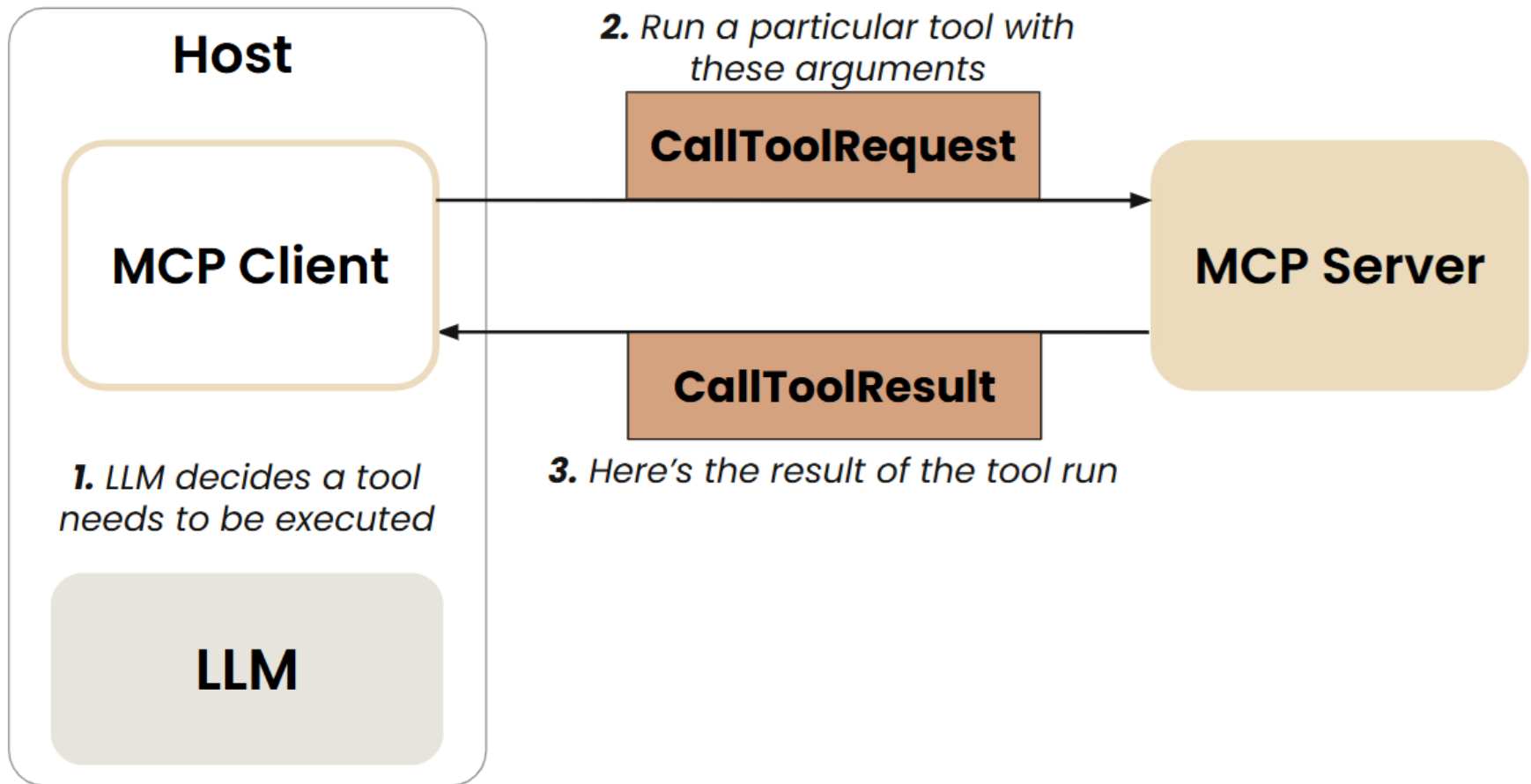
MCP parts

- MCP Client
 - AI application requiring data or tools (e.g. chatbot or agent)
- MCP Server
 - Server that exposes tools, data, or services that client can utilize
- Support local operation via `stdin/stdout` or remote operation via HTTP or WebSockets



Details





MCP reference servers

- <https://github.com/modelcontextprotocol/servers>
 - Fetch (web content)
 - Filesystem
 - [GitHub](#)
 - [AWS](#)
 - [Terraform](#)
 - Search (Brave, Google)
 - Google Maps
 - Google Drive
 - Slack
 - Sqlite/PostgreSQL/Chroma
 - Google Cloud Run, Google Calendar

- [Playwright](#)



Debbie O'Brien

Posted on Jun 18



Letting Playwright MCP Explore your site and Write your Tests

#mcp #playwright #ai #testing

Example: Vulnerable SQLite MCP server

- FastMCP server to allow an agent to query a database
 - What's wrong with this code?
 - Insecure tool design will keep us all employable

```
from fastmcp import FastMCP
import sqlite3

mcp = FastMCP("sqlite")
con = sqlite3.connect('db_data/metactf_users.db')

@mcp.tool()
async def query(query: str) -> list:
    """Query a Sqlite3 database. Takes a query
    string as an input parameter and returns the result of
    the query."""
    cur = con.cursor()
    res = cur.execute(query)
    con.commit()
    return res.fetchall()
```

- MCP client agent

```
from langchain_mcp_adapters.tools import load_mcp_tools
from mcp import ClientSession
from mcp.client.stdio import StdioServerParameters, stdio_client
server = StdioServerParameters(
    command="python",
    args=["vulnerable_sqlite_mcp_server.py", "stdio"]
)
```

prompt = f"You are a Sqlite3 database look up tool. Perform queries on a database hosted by the specified MCP server given the user's input. Utilize the user input verbatim when sending the query to the database and print the query that was sent to the database"

```
async def run_agent():
    async with stdio_client(server) as (read, write):
        async with ClientSession(read, write) as session:
            await session.initialize()
            tools = await load_mcp_tools(session)
            agent = create_agent(model=llm, tools=tools,
system_prompt=prompt, debug=True)
            query = "Who are the users in the database?"
            result = await agent.ainvoke({"messages": [("user", query)]})
            print(f"Agent response: {result}")
```

Labs

- Custom MCP servers for
 - Google Drive
 - Git
 - Fetch
 - Curl

Homework

- Custom MCP application
 - One MCP server of your own
 - One MCP server integrated from <https://github.com/modelcontextprotocol/servers>
- MCP client agent that can solve a multi-step prompt utilizing tools from each server